Project Report: End-to-End Sales Forecasting & Demand Intelligence System

Week 3 & 4 Internship Project Assignment

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1. Executive Summary

We have conducted a comprehensive review of $2.26M in historical retail sales data to establish a data-driven sales forecasting and inventory optimization framework. By linking statistical seasonality models with machine learning anomaly detection and product demand clustering, this system provides supply chain leaders with the tools to preemptively align inventory levels with customer demands. The core insights allow the company to free up tied-up working capital in high-value machinery through shift-to-order models, automate restocking schedules for high-turnover office basics, and secure carrier freight contracts in advance of forecasted demand spikes, protecting operational margins and service levels.

2. Key Findings from Historical Data & Modeling

- Predictable Annual Seasonality: Sales display strong, recurring holiday surges in Q4 (specifically November and December) driven by consumer holiday shopping, followed by sharp volume drops of up to 45% in January and February as corporate budgets reset.
- Institutional B2B Spikes: The highest weekly sales volumes are not driven by steady retail purchases but by isolated, high-value corporate B2B contracts (specifically in Machines and Copiers), which create massive volatility in cash flow.
- SARIMA Model Superiority: During testing against historical records, the Seasonal AutoRegressive Integrated Moving Average (SARIMA) model emerged as our best planning baseline, achieving a low error rate (11.45% MAPE) that significantly outperformed Facebook Prophet (21.89% MAPE) and XGBoost (32.78% MAPE).

3. Three-Month Sales Forecast & Supply Chain Capacity Plan

Using our SARIMA forecasting model, we have projected total corporate sales for the upcoming first quarter of 2019. The model predicts a quiet post-holiday winter baseline followed by an aggressive, high-probability spring recovery in March.Below are the specific expected monthly sales totals and their statistically realistic operating ranges (95% confidence bounds) to guide financial budgeting and warehouse scheduling:

Future Month	Expected Sales Forecast ($)	Realistic Operating Range (95% Confidence)
January 2019	$46,782.36	$16,992.55 to $76,572.16
February 2019	$40,285.41	$9,595.90 to $70,974.92
March 2019	$72,234.21	$41,318.78 to $103,149.64

4. Historical Anomaly Analysis & Root Causes

Usingmachinelearning (IsolationForest), we analyzedweeklytransaction history to isolate irregular outliers from normal seasonal behavior. The table below details the top 3 largest anomalies and their underlying business drivers:

Week Ending	Weekly Sales ($)	Primary Product Category Driver	Likely Cause & Context
2015-03-22	$37,703.67	Technology: Machines ($24,739.75 - 66% of total)	Large-scale corporate B2B bulk purchase or major institutional hardware refresh contract.
2018-12-02	$35,998.90	Technology: Phones ($10,981.27 - 31% of total)	Black Friday & Cyber Monday holiday consumer sales surge exceeding regular baseline targets.
2018-11-18	$30,572.45	Technology: Copiers ($11,399.95 - 37% of total)	End-of-year corporate B2B copier upgrade cycle matching typical Q4 budget clearance behavior.

5. Product Demand Segmentation & Stocking Strategy

Demand Cluster	Categories Included	Stocking Policy	Operational & Financial Rationale
Cluster 0: High-Value, High-Volatility	Copiers, Machines	Just-in-Time (JIT) / Pull Procurement	Avoid tying up working capital in expensive, slow-moving units. Coordinate direct drop-shipping contracts with vendors.
Cluster 1: High-Volume, High-Stability	Binders, Paper, Phones, Storage, Furnishings	Fixed Reorder Point (ROP) Replenishment	Core sales drivers. Maintain high buffer stocks with automatic reorder triggers based on EOQ logic to prevent stockouts.
Cluster 2: Moderate-Volume, Low-Volatility	Chairs, Tables, Appliances, Envelopes, Art, etc.	Periodic Review (Min-Max Checking)	Review stock levels monthly. Balance warehousing footprint and holding costs against bulk carrier transportation freight fees.

6. Concrete Business Recommendations

1.Transition Segment 0 Capital Goods to JIT Vendor Fulfillment

Data Backing: High-cost items (Machines/Copiers) drive irregular revenue spikes, such as the single weekly Machines contract of **$24,739.75 on March 22, 2015** (66% of that week's sales). Holding this inventory on the warehouse floor ties up valuable working capital and carries depreciation risks. **Action:** Establish direct vendor drop-shipping agreements to fulfill these large corporate contracts on demand, removing them completely from warehouse floor footprints.

2. Automate Replenishment for Segment 1 Everyday Staples

Data Backing: High-turnover office staples (Paper, Binders, and Phones) represent consistent baseline transaction volumes (e.g. Phones hit **$10,981.27** during the Cyber Week of December 2, 2018). **Action:** Transition these products to an automated inventory control system using calculated Reorder Points (ROP). This reduces administrative ordering overhead and eliminates the risk of stockouts during major demand cycles.

3. Pre-Book Carrier Logistics and Labor for the March Demand Spike

Data Backing: Our forecasting model projects that monthly sales will surge by **79.3%** from February ($40,285.41) to March ($72,234.21). **Action:** The Head of Supply Chain should pre-book truck freight carrier capacity and increase warehouse temp labor allocations by mid-February to prevent shipping bottlenecks and freight rate premiums.

7. Analytical System Risk & Limitation

Reliance on Historical Continuity: Theforecastingengine is heavily dependent on historical seasonal patterns repeating. It cannot anticipate structural market changes, macro-economic disruptions, supplier shutdowns, or major changes in corporate purchasing rules (e.g., a policy switch away from high-cost copiers). Consequently, these models must serve as a planning baseline to assist supply chain managers, rather than a completely autonomous procurement decision-maker.